

Handout 1.6: Silent Measurement Challenge Cards

Print one card per team (four teams, cards A to D) and one teacher scoring sheet per class. Cut the cards apart. Each team also needs six paper cups or small bowls labeled 1 to 6, the measuring tools, and the five ingredients.

The rules (print on the back of every card)

1. No talking. No writing notes to each other. No mouthing words.
2. You may point, gesture, hold up fingers, and show the card.
3. One person measures each item. A different person measures each cup. Everyone else watches and signals.
4. Each cup gets one measurement. Once a cup is set down on the tray, it is done. Do not touch it again.
5. Use the right tool: dry cups for flour and sugar, the glass cup for water and oil, spoons for small amounts.
6. When all six cups are on the tray, carry the tray to the weigh-in table together.

Points per cup: 3 in the tight band, 2 in the wide band, 1 for the right ingredient with the wrong amount, 0 for the wrong ingredient or an empty cup. 18 points possible.



Team Card A

Team name: _____

Cup	Measure this
1	1/2 cup flour
2	1/4 cup sugar
3	1 cup water
4	2 tablespoons oil
5	1 teaspoon salt
6	3 teaspoons water

Grade 8 stretch cup 7: 1 1/2 tablespoons oil.



Team Card B

Team name: _____

Cup	Measure this
1	1/4 cup flour
2	1/2 cup sugar
3	1/2 cup water
4	1 tablespoon oil
5	2 teaspoons salt
6	4 tablespoons water

Grade 8 stretch cup 7: 3/4 cup water using only the 1/4 cup and the 1/2 cup dry measures.



Team Card C

Team name: _____

Cup	Measure this
1	1 cup flour
2	2 tablespoons sugar
3	3/4 cup water
4	1/4 cup oil
5	1/2 teaspoon salt
6	1 tablespoon flour

Grade 8 stretch cup 7: 1 1/2 tablespoons oil.



Team Card D

Team name: _____

Cup	Measure this
1	$\frac{3}{4}$ cup flour
2	$\frac{1}{3}$ cup sugar
3	$\frac{1}{4}$ cup water
4	3 tablespoons oil
5	1 tablespoon salt
6	2 tablespoons water

Grade 8 stretch cup 7: $\frac{3}{4}$ cup water using only the $\frac{1}{4}$ cup and the $\frac{1}{2}$ cup dry measures.

Teacher scoring sheet

Class: _____ Date: _____

Typical targets use these weights: water 237 g per cup (15 g per Tbsp, 5 g per tsp); flour, spooned and leveled, 120 g per cup (about 8 g per Tbsp); granulated sugar 200 g per cup (about 12.5 g per Tbsp); vegetable oil about 13.6 g per Tbsp (about 218 g per cup); table salt about 6 g per tsp. Tight band is within 5 percent or 2 g, whichever is larger. Wide band is within 12 percent or 4 g, whichever is larger. *[Sal: weigh each item with the room's own cups and flour the day before and write your numbers in the "my target" column; flour can run 10 percent heavy on a humid day. Tare the scale on an empty cup before each weigh-in.]*

Card A

Cup	Item	Typical target (g)	My target (g)	Tight (3 pts)	Wide (2 pts)	Team reading (g)	Points
1	1/2 c flour	60		57 to 63	53 to 67		
2	1/4 c sugar	50		47 to 53	44 to 56		
3	1 c water	237		225 to 249	209 to 265		
4	2 Tbsp oil	27		25 to 29	23 to 31		
5	1 tsp salt	6		4 to 8	2 to 10		
6	3 tsp water (1 Tbsp)	15		13 to 17	11 to 19		
7 (stretch)	1 1/2 Tbsp oil	20		18 to 22	16 to 24		
						Card A total (18)	

Card B

Cup	Item	Typical target (g)	My target (g)	Tight (3 pts)	Wide (2 pts)	Team reading (g)	Points
1	1/4 c flour	30		28 to 32	26 to 34		



2	1/2 c sugar	100		95 to 105	88 to 112		
3	1/2 c water	118		112 to 124	104 to 132		
4	1 Tbsp oil	14		12 to 16	10 to 18		
5	2 tsp salt	12		10 to 14	8 to 16		
6	4 Tbsp water (1/4 c)	59		56 to 62	52 to 66		
7 (stretch)	3/4 c water by 1/4 + 1/2	178		169 to 187	157 to 199		
						Card B total (18)	

Card C

Cup	Item	Typical target (g)	My target (g)	Tight (3 pts)	Wide (2 pts)	Team reading (g)	Points
1	1 c flour	120		114 to 126	106 to 134		
2	2 Tbsp sugar	25		23 to 27	21 to 29		
3	3/4 c water	178		169 to 187	157 to 199		
4	1/4 c oil	55		52 to 58	48 to 62		
5	1/2 tsp salt	3		1 to 5	0.5 to 7		
6	1 Tbsp flour	8		6 to 10	4 to 12		
7 (stretch)	1 1/2 Tbsp oil	20		18 to 22	16 to 24		
						Card C total (18)	

Card D

Cup	Item	Typical target (g)	My target (g)	Tight (3 pts)	Wide (2 pts)	Team reading (g)	Points
1	3/4 c flour	90		85 to 95	79 to 101		
2	1/3 c sugar	67		64 to 70	59 to 75		
3	1/4 c water	59		56 to 62	52 to 66		
4	3 Tbsp oil	41		39 to 43	36 to 46		
5	1 Tbsp salt	18		16 to 20	14 to 22		
6	2 Tbsp water	30		28 to 32	26 to 34		
7 (stretch)	3/4 c water by 1/4 + 1/2	178		169 to 187	157 to 199		
						Card D total (18)	

Scoreboard

Team	Card	Total (18)	Cups in the tight band	The one miss to talk about in the debrief
	A			
	B			
	C			
	D			

What to watch for during the round (one note per team for the debrief):

Team	How they handled a disagreement without words	The signal that worked
A		

B		
C		
D		

Reading the misses: flour heavy is packing or scooping; flour light is the liquid cup read from above, or a heavy tap that settled it; sugar off is almost always not leveled; water off is reading from above (heavy) or a spill on the way to the cup (light); oil light is the film left in the spoon; salt heavy is a heaped spoon.

Grade 6 version: use cups and whole spoons only. Change Card C cup 5 to 1 tsp salt (6 g), Card D cup 1 to 1/2 c flour (60 g), and Card D cup 2 to 1/4 c sugar (50 g); drop the stretch cups. Still 18 points.